

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T47 · Area and Volume of Similar Shapes

Questions

Student book · questions with writing space

■ 12 pages

FAMILY MAP

Area Volume of Similar Shapes

F01	Use volume scale factors of similar solids	3 Q 11 marks
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F02	Link length, area and volume similarity	5 Q 21 marks
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F03	Use area scale factors of similar figures	3 Q 10 marks
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26 **A** and **B** are two similar solids.

The height of solid **A** is 31 cm

The height of solid **B** is 18.6 cm

Given that

$$\text{volume of solid A} - \text{volume of solid B} = 735 \text{ cm}^3$$

find the volume of solid **A**

..... cm³

(Total for Question 26 is 4 marks)

Question 18

4MA1-1H Q23 demand

MODULAR UNIT 2

5 marks

23 Solid **A** is similar to solid **B**

Here is some information about solid **A** and solid **B**

	solid A	solid B
Height (cm)	3^x	
Area (cm ²)	7776	486
Volume (cm ³)	8^x	2^{x+4}

Work out the height of solid **B**

Give your answer as a decimal.

..... cm

Question 36

4MA1-2H Q23 whole

MODULAR UNIT 2

5 marks

23 Here is a frustum of a cone.

The frustum is made by removing a small cone from a similar large cone.

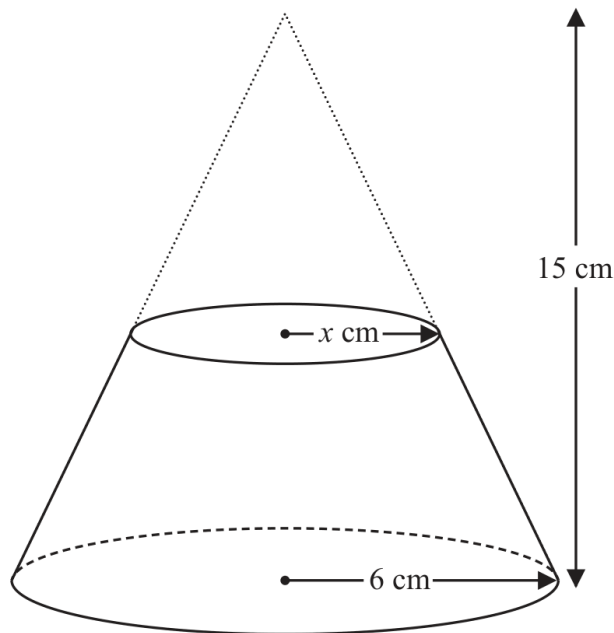


Diagram **NOT**
accurately drawn

The height of the large cone is 15 cm.

The radius of the base of the large cone is 6 cm.

The radius of the base of the small cone is x cm.

Given that the volume of the frustum is $\frac{4212}{25}\pi \text{ cm}^3$

work out the value of x

Show clear algebraic working.

$x = \dots\dots\dots$

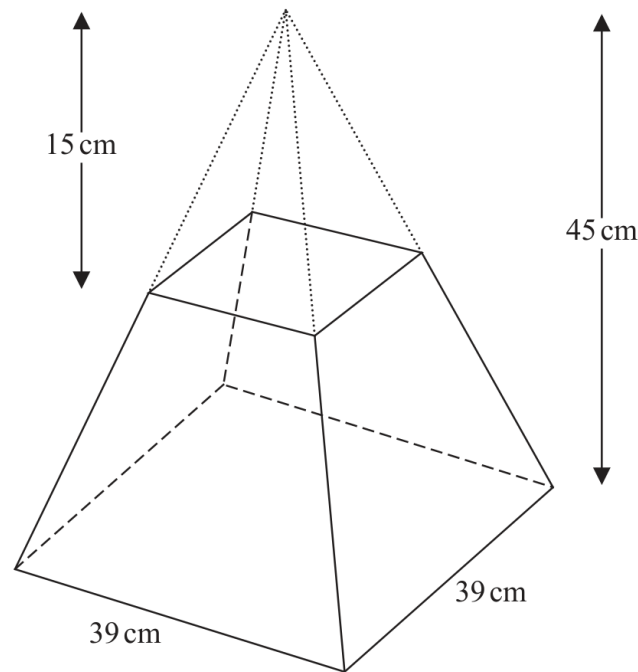
Question 18

4MA1-1H Q23

MODULAR UNIT 2

5 marks

- 23 A frustum is made by removing a small square-based pyramid from a similar large squared-based pyramid as shown in the diagram.



The height of the small pyramid is 15 cm.

The height of the large pyramid is 45 cm.

The square base of the large pyramid has side length 39 cm.

Work out the **total** surface area of the frustum.

Give your answer correct to the nearest whole number.

..... cm²

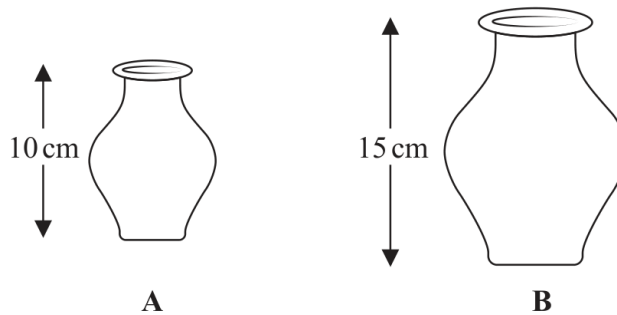
Question 11

4MA1-1H Q17

MODULAR UNIT 2

4 marks

17 **A** and **B** are two similar vases.



Vase **A** has height 10 cm.

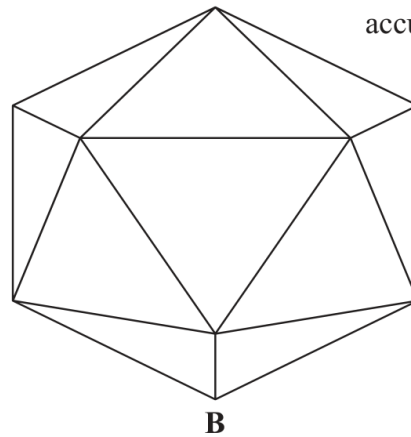
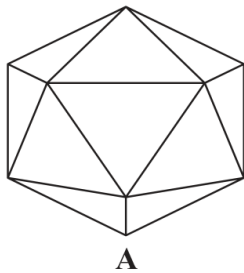
Vase **B** has height 15 cm.

The difference between the volume of vase **A** and the volume of vase **B** is 1197 cm^3

Calculate the volume of vase **A**

..... cm^3

Diagram **NOT**
accurately drawn



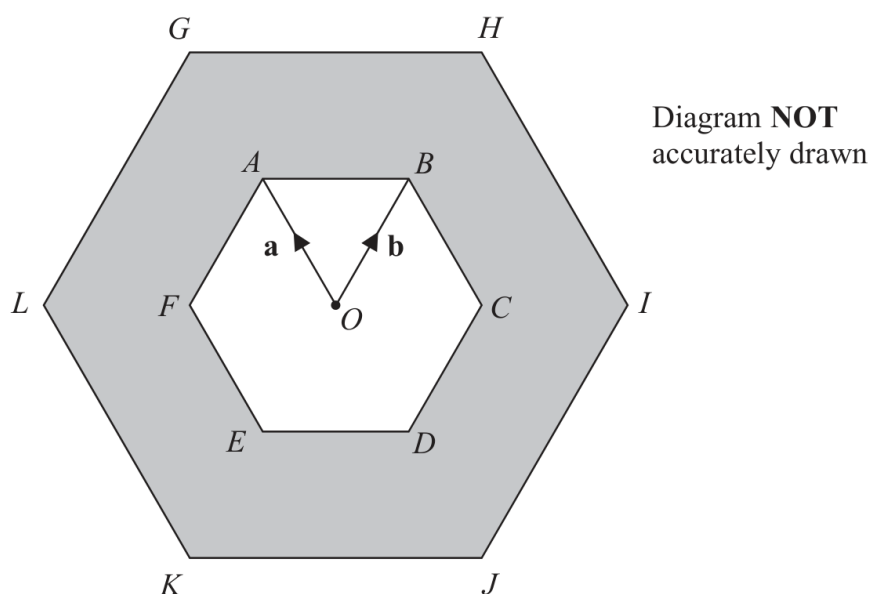
B has a volume of 4352 cm^3

Work out the total surface area of **A**.

..... cm²

..... cm³

14 $ABCDEF$ and $GHIJKL$ are regular hexagons each with centre O .



$GHIJKL$ is an enlargement of $ABCDEF$, with centre O and scale factor 2

$$\vec{OA} = \mathbf{a} \quad \vec{OB} = \mathbf{b}$$

(a) Write the following vectors, in terms of $\mathbf{a}$ and $\mathbf{b}$.
Simplify your answers.

(i) $\vec{AB}$

(1)

(ii) $\vec{KI}$

(2)

(iii) $\vec{LD}$

(2)

The triangle OAB has an area of 5 cm^2

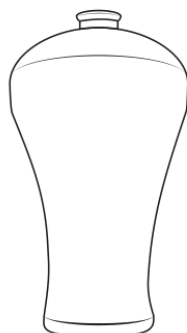
(b) Calculate the area of the shaded region.

cm^2

..... cm³

(Total for Question 20 is 3 marks)

19 **A** and **B** are two similar vases.



A



B

Diagram **NOT**
accurately drawn

The vases are such that

$$\text{surface area of vase B} = \frac{25}{64} \times \text{surface area of vase A}$$

and that

$$\text{volume of vase A} - \text{volume of vase B} = 541.8 \text{ cm}^3$$

Calculate the volume of vase **B**

..... cm^3

23 Shape P is similar to shape Q

The table shows some information about shape P and shape Q

	Surface area (cm ²)	Volume (cm ³)
Shape P	200	672
Shape Q	450	

Work out the volume of shape Q

..... cm³